

Library Management

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2024-25

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Chapter 1

Introduction

This project presents a Library Management System developed using Python for the frontend and SQL for the backend. The primary objective of this system is to simplify and streamline the process of managing a library's operations, including the cataloging of books, tracking issued books and managing returns.

The Python-based frontend ensures a user-friendly interface, making it easy for library staff and users to interact with the system. On the other hand, SQL powers the backend, ensuring robust data storage and management. This system integrates key functionalities such as adding new books, updating existing entries, issuing books to members, and returning them, while maintaining an efficient database to track transactions and book availability.

By leveraging Python's versatility and SQL's powerful data management capabilities, this Library Management System offers a comprehensive solution for libraries, aiming to reduce manual effort, improve accuracy, and enhance the overall experience librarians and library users.

1.1 Features

1.1.1 Filters

Filters are incorporated into the program to browse the library. These filters insure a smooth experience for the user, to find the right book for them.

ISBN

This feature allows to search for the books via the International Standard Book Number (ISBN). This is unique for every book, and hence is also set as the primary key. This feature reduces redundancy in the data and allows for easier searching.

Book Name

This function allows you to search by the name of the book. This function is incorporated in the case of unavailability of the ISBN. And there is a choice of searching via the ISBN or the name of the book.

Author

This function allows to filter *all* the books by an author.

Tags

This feature allows to filter the books via certain tags, allowing for easier browsing of particular genres.

Year

This feature allows to filter the books in a range of years that they were added to the library.

1.1.2 Functions

Add Book

This function allows you to add the books to the library.

Remove Book

This function allows to remove the books from the library.

Issue Book

This function allows to issue a book from the library.

Return Book

This function allows us to return the issued book back to the library.

Chapter 2

Examples and Outputs

Below are some example screenshots that demonstrate the key features and functionality of the Library Management System. These images show the user interface and how different operations are performed within the system, providing a visual representation of the interaction between the front-end and the back-end. These screenshots offer a clear overview of how users can navigate through the system and perform essential tasks with ease.

2.1 Home Screen

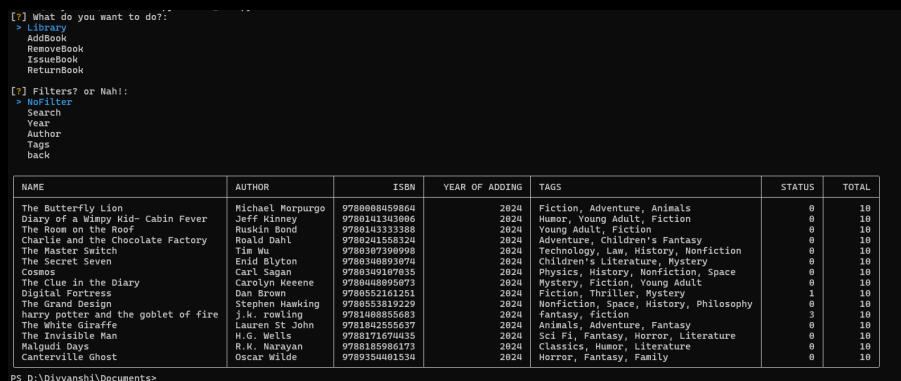
Displays the main dashboard with navigation options for different functionalities such as managing books, members, and transactions.

2.2 Library

Is the large catalog of books which can be browsed using author's name, ISBN, book name etc.

2.2.1 No Filter

This filter helps in just browsing the library without any restrictions.



NAME	AUTHOR	ISBN	YEAR OF ADDING	TAGS	STATUS	TOTAL
The Butterfly Lion	Michael Morpurgo	97808000459864	2024	Fiction, Adventure, Animals	0	10
Diary of a Wimpy Kid- Cabin Fever	Jeff Kinney	9780141343006	2024	Humor, Young Adult, Fiction	0	10
The Room on the Roof	Ruskin Bond	9780143333388	2024	Young Adult, Fiction	0	10
Charlie and the Chocolate Factory	Roald Dahl	9780203550324	2024	Adventure, Children's Fantasy	0	10
The Master Switch	Tim Wu	9780307390998	2024	Technology, Law, History, Nonfiction	0	10
The Secret Seven	Enid Blyton	9780340893074	2024	Children's Literature, Mystery	0	10
Cosmos	Carl Sagan	9780340107035	2024	Physics, History, Nonfiction, Space	0	10
The Clue in the Diary	Carolyn Keene	9780440895073	2024	Mystery, Fiction, Young Adult	0	10
Digital Fortress	Dan Brown	9780552161251	2024	Fiction, Thriller, Mystery	1	10
The Grand Design	Stephen Hawking	9780553810229	2024	Nonfiction, Space, History, Philosophy	0	10
harry potter and the goblet of fire	j.k. Rowling	9781408855683	2024	Fantasy, Fiction	3	10
The white Giraffe	Lauren St John	9781842555637	2024	Animals, Adventure, Fantasy	0	10
The Invisible Man	H.G. Wells	9780171670435	2024	Sci Fi, Fantasy, Horror, Literature	0	10
Malgudi Days	R.K. Narayan	9780185986173	2024	Classics, Humor, Literature	0	10
Canterville ghost	Oscar Wilde	9789354401534	2024	Horror, Fantasy, Family	0	10

2.2.2 Tags

This helps the user find a particular kind of books pertaining to one's interests and taste.

```
PS D:\Divyanshi\Downloads> python IP_PRJ.py
[?] What do you want to do?:
> Library
  AddBook
  RemoveBook
  IssueBook
  ReturnBook

[?] Filters? or Nah!:
  NoFilter
  Search
  Year
  Author
> Tags
  back

[?] Are you sure?:
> Continue
  back

Enter the tags separated by comma : Fiction
```

NAME	AUTHOR	ISBN	YEAR OF ADDING	TAGS	STATUS	TOTAL
The Butterfly Lion	Michael Morpurgo	97800008459864	2024	Fiction, Adventure, Animals	0	10
Diary of a Wimpy Kid- Cabin Fever	Jeff Kinney	9780141343086	2024	Humor, Young Adult, Fiction	0	10
The Room on the Roof	Ruskin Bond	9780143333388	2024	Young Adult, Fiction	0	10
The Master Switch	Tim Wu	9780307390998	2024	Technology, Law, History, Nonfiction	0	10
Cosmos	Carl Sagan	9780349107035	2024	Physics, History, Nonfiction, Space	0	10
The Clue in the Diary	Carolyn Keene	9780440095073	2024	Mystery, Fiction, Young Adult	0	10
Digital Fortress	Dan Brown	9780552161251	2024	Fiction, Thriller, Mystery	0	10
The Grand Design	Stephen Hawking	9780553819229	2024	Nonfiction, Space, History, Philosophy	0	10
The Age of Miracles	Karen Thompson Walker	9780812992977	2024	Coming of Age, Fantasy, Fiction	0	10

```
PS D:\Divyanshi\Downloads>
```

2.2.3 Author

This filter *filters* the books based off of a particular author.

```
PS D:\Divyanshi\Documents> python IP_PRJ.py
[?] What do you want to do?:
> Library
  AddBook
  RemoveBook
  IssueBook
  ReturnBook

[?] Filters? or Nah!:
  NoFilter
  Search
  Year
> Author
  Tags
  back

[?] Are you sure?:
> Continue
  back

Enter the Author of the Book : Tim Wu
```

NAME	AUTHOR	ISBN	YEAR OF ADDING	TAGS	STATUS	TOTAL
The Master Switch	Tim Wu	9780307390998	2024	Technology, Law, History, Nonfiction	0	10

```
PS D:\Divyanshi\Documents>
```

2.3 Add New Book

Shows the interface where new books can be added to the system's database by entering relevant details like title, author, genre, and publication date.

```

PS D:\Divyanshi\Documents> python IP_PRJ.py
[?] What do you want to do?:
Library
> AddBook
RemoveBook
IssueBook
ReturnBook

Enter the SchoolID:: 5237
Enter the name of the book : Harry Potter and the Goblet of Fire
Enter the name of the author(s) (separated by comma) : J.K. Rowling
Enter the ISBN of the book : 9781408855683
Enter the year of adding of the book(YYYY) : 2024
Enter the tags of the book(separated by comma) : Fantasy, Fiction
Enter the number of books lent : 3
Enter the total number of books : 10
harry potter and the goblet of fire Added.
PS D:\Divyanshi\Documents>

```

2.4 Issue Book

Illustrates the process of issuing a book to a registered library member, highlighting fields such as book ID, member ID, and issue date.

```

PS D:\Divyanshi\Downloads> python IP_PRJ.py
[?] What do you want to do?:
Library
AddBook
RemoveBook
> IssueBook
ReturnBook

Enter the SchoolID of the borrower : 5237
[?] Search By::
> BookName
ISBN
back

[?] Are you sure?:
> Continue
back

Enter the name of the Book : Digital Fortress

```

NAME	AUTHOR	ISBN	YEAR OF ADDING	TAGS	STATUS	TOTAL
Digital Fortress	Dan Brown	9780552161251	2024	Fiction, Thriller, Mystery	0	10

```

Digital Fortress Issued
PS D:\Divyanshi\Downloads>

```

2.5 Return Book

Depicts the interface for managing book returns, including verifying the return date and updating the status of the book in the system.

2.6 Delete Book

Depicts the interface for deleting the books which have come of age.

```
PS D:\Divyanshi\Documents> python IP_PRJ.py
[?] What do you want to do?:
Library
AddBook
RemoveBook
IssueBook
> ReturnBook
```

Enter the SchoolID of the person : 5237

```
[?] Search By::
> BookName
ISBN
back
```

[?] Are you sure?:

```
> Continue
back
```

Enter the name of the Book : Cosmos

NAME	AUTHOR	ISBN	YEAR OF ADDING	TAGS	STATUS	TOTAL
Cosmos	Carl Sagan	9780349107035	2024	Physics, History, Nonfiction, Space	1	10

Cosmos Returned

PS D:\Divyanshi\Documents>

```
PS D:\Divyanshi\Downloads> python IP_PRJ.py
[?] What do you want to do?:
Library
AddBook
> RemoveBook
IssueBook
ReturnBook
```

Enter the SchoolID:: 5237

```
[?] Search By::
> BookName
ISBN
back
```

[?] Are you sure?:

```
> Continue
back
```

Enter the name of the Book : The Age of Miracles

NAME	AUTHOR	ISBN	YEAR OF ADDING	TAGS	STATUS	TOTAL
The Age of Miracles	Karen Thompson Walker	9780812992977	2024	Coming of Age, Fantasy, Fiction	0	10

1 record(s) deleted

PS D:\Divyanshi\Downloads>

2.7 Logs

This table depicts the logs that will be registered according to the time of action.

```
mysql> select * from log;
```

ACTION	BOOK_ISBN	STUDENT_ID	TIME
ISSUED	9780349107035	5237	2024-11-12 14:30:47
ISSUED	9780812992977	5237	2024-11-12 14:32:09
ISSUED	9780552161251	5237	2024-11-12 14:40:12
RETURNED	9780349107035	5237	2024-11-12 14:45:54
ADDED	9781408855683	5237	2024-11-12 15:20:10
ISSUED	9781408855683	5237	2024-11-12 15:21:39
ADDED	9781408855683	5237	2024-11-12 15:23:02

7 rows in set (0.00 sec)

```
mysql>
```


Chapter 3

Bibliography

Idea for a CLI:

<https://youtu.be/20Qkq93kwKw?feature=shared>

Extra modules Used:

Typer: <https://typer.tiangolo.com/>

Tabulate: https://pyneng.readthedocs.io/en/latest/book/12_useful_modules/tabulate.html

Inquirer: <https://python-inquirer.readthedocs.io/en/latest/index.html>

A major thanks to:

[Stackoverflow](#)

[GeeksforGeeks](#)

Books:

Informatics Practices by Sumita Arora